Pinned group summary: `SO(n=6, q=1)`

Pinning-Sympy automated output

July 6, 2026

Contents

1 Basic group attributes

Group	`SO(n=6, q=1)`
Matrix size	6×6
Rank	1
Defining equations	$DefiningEquationPlaceholder$

1.1 Symmetric bilinear form

Dimension	6
Witt index	1
Primitive element	N/A
Epsilon	N/A

Matrix: $\begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & c_0 & 0 & 0 & 0 \\ 0 & 0 & 0 & c_1 & 0 & 0 \\ 0 & 0 & 0 & 0 & c_2 & 0 \\ 0 & 0 & 0 & 0 & 0 & c_3 \end{bmatrix}$

1.2 Generic Lie algebra element

$$\begin{bmatrix} x_{00} & 0 & -c_0x_{21} & -c_1x_{31} & -c_2x_{41} & -c_3x_{51} \\ 0 & -x_{00} & -c_0x_{20} & -c_1x_{30} & -c_2x_{40} & -c_3x_{50} \\ x_{20} & x_{21} & 0 & x_{23} & x_{24} & x_{25} \\ x_{30} & x_{31} & -\frac{c_0x_{23}}{c_1} & 0 & x_{34} & x_{35} \\ x_{40} & x_{41} & -\frac{c_0x_{24}}{c_2} & -\frac{c_1x_{34}}{c_2} & 0 & x_{45} \\ x_{50} & x_{51} & -\frac{c_0x_{25}}{c_3} & -\frac{c_1x_{35}}{c_3} & -\frac{c_2x_{45}}{c_3} & 0 \end{bmatrix}$$

1.3 Generic torus element

$$\begin{bmatrix} t_0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{t_0} & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

1.4 Trivial characters

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

2 Root system

Summary Properties

Dynkin type	`A1`
Irreducible	True
Reduced	True
Simply laced	True
Number of roots	2

2.1 Dynkin diagram

0

2.2 Cartan matrix

$$\begin{bmatrix} 2 \end{bmatrix}$$

2.3 Roots and coroots

Root	**Coroot**	**Simple**	**Sign**	**Multipliable**
$(-1,\ 0,\ 0,\ 0,\ 0,\ 0)$	$(-2,\ 2,\ 0,\ 0,\ 0,\ 0)$	No	$-$	No
$(1,\ 0,\ 0,\ 0,\ 0,\ 0)$	$(2,\ -2,\ 0,\ 0,\ 0,\ 0)$	Yes	$+$	No

2.4 Linear combinations of roots

Definition 2.1. For two roots α, β in a root system Φ, the set of positive integral linear combinations of α, β that are also roots is denoted $(\alpha, \beta) = \{i\alpha + j\beta \in \Phi : i, j \in \mathbb{Z}_{\geq 1}\}$

No pairs of roots generate positive integer linear combinations.

3 Root spaces

Root	Dimension
$(-1,\ 0,\ 0,\ 0,\ 0,\ 0)$	4
$(1,\ 0,\ 0,\ 0,\ 0,\ 0)$	4

RootSpaceTablePlaceholder

4 Root subgroups

RootSubgroupTablePlaceholder
HomomorphismDefectTablePlaceholder
HomomorphismDefectIdentitiesPlaceholder

5 Commutators

CommutatorCoefficientTablePlaceholder
CommutatorIdentitiesPlaceholder

6 Weyl group

WeylGroupPlaceholder
WeylConjugationCoefficientPlaceholder
WeylConjugationIdentitiesPlaceholder

7 Coroot torus elements (h_α)

CorootTorusElementPlaceholder